Muhammad Farrell Astrada Azhar

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Biomedical Engineering graduate with research experience in biosignal processing, wearable sensing, embedded machine learning, and medical image analysis, complemented by hands-on experience in medical-device R&D and regulatory documentation.

Education

Institut Teknologi Bandung (itb.ac.id) *Aug 2021 - Sep 2026*
Bachelor of Engineering in Biomedical Engineering

Research and Technical Projects

Development of a Wearable Sensor System for the Detection and Classification of Cognitive and Physiological Stress Using Photoplethysmograph, Electrodermal Activity, and Accelerometer Sensors (Undergraduate Thesis) *Sep 2025 – Sep 2026*

- Developed an ESP32-S3-based multimodal wearable integrating PPG (MAX30102), EDA/GSR, and accelerometry (MPU6050) to distinguish baseline, cognitive stress, and aerobic activity conditions.
- Developed a 57-feature Random Forest classifier evaluated using Leave-One-Subject-Out Cross-Validation (LOSO-CV) across 23 subjects, achieving 86.41% accuracy, 85.55% balanced accuracy, and 83.92% macro-F1.
- Deployed the end-to-end signal-processing and ML pipeline on the ESP32-S3 using FreeRTOS, including 100 Hz acquisition, digital filtering, 60-s windowing, feature extraction, personalized baseline normalization, and on-device inference.
- Achieved 99.35% prediction agreement between the Python and ESP32-S3 implementations, with BLE communication to a Python GUI for real-time monitoring and data logging.

BITalino Biosignal Reader *Aug 2026*

- Developed a Python desktop application for real-time BITalino biosignal acquisition and visualization, including EDA/SCL/SCR processing, CSV recording, BLE communication, and simulation-based testing. [GitHub]

Segmentation of Exudates in Fundus Images for Diabetic Retinopathy *Feb 2025 – Jul 2025*

- Developed a multi-stage retinal image-processing pipeline using Lanczos resizing, CLAHE, Unsharp Masking, Renyi entropy thresholding, and quantitative feature extraction for exudate segmentation.
- Compared SVM, Random Forest, KNN, and Logistic Regression classifiers; the selected SVM achieved 86% F1-score, 87% precision, and 86% recall. Listed as a co-creator under Copyright Registration No. 000938547. [Certificate]

Smart Hand Stroke Rehabilitation Device (*Capstone Design Project*) *Jan 2024 – Dec 2024*

- Conducted system-level specification analysis and component trade-off studies for an ESP32-based stroke rehabilitation device, including hardware selection and system architecture design.
- Developed an Android application using MediaPipe for real-time hand-gesture tracking and integrated BLE communication with the ESP32-based rehabilitation device for motion control.

Professional Experience

PT. Forsta Kalmedic Global
Research and Development Intern *Jun 2024 – Aug 2024*

- Assisted in drafting 10+ technical and regulatory documents, including SOPs, Clinical Evaluation Reports, and Quality Risk Management documentation, referencing ISO 14971:2019 and ISO/TR 24971:2020 risk-management guidance.
- Supported an early-stage PACS development process and gained introductory exposure to IEC 62304 requirements for the medical-device software lifecycle.
- Analyzed clinical workflows and conducted dose-optimization studies for X-ray systems to support patient safety and operational effectiveness.

RSUD Chasbullah Abdul Madjid Kota Bekasi
Biomedical Equipment Technician Intern *Jan 2024 – Feb 2024*

- Analyzed and documented equipment maintenance workflows, identifying inefficiencies and providing troubleshooting recommendations.
- Conducted troubleshooting and component-level repair to restore functionality in compliance with safety and performance standards.

Justisio
Digital Marketing Staff *Mar 2025 – Jun 2025*

- Built an automated Gemini API workflow that processed curated news links and generated 10–15 content ideas daily, using structured prompting and spreadsheet integration for scheduled content planning.

Organizational Experience

Himpunan Mahasiswa Elektroteknik ITB
Staff, Career Preparation Division *Sep 2022 – Feb 2024*

- Organized workshops, scientific reviews, and industry visits, and developed an open-source career roadmap to support students' career preparation.

SatuEdu Foundation (formerly Skill Up ID)
MC & Moderator Specialist *Nov 2021 – Apr 2022*

- Hosted and moderated 10+ professional events with 1,500+ total attendees and 25+ speakers.

Certifications

Google Cloud Computing Foundations: Cloud Computing Fundamentals *Sep 2025*
Google Cloud Computing Foundations: Infrastructure in Google Cloud *Sept2025*
Dicoding Fundamentals of Data Visualization *Sep 2024*

Skills

Technical: Python, C++, Kotlin, MATLAB, machine learning, biosignal processing, digital signal processing, medical image processing, embedded systems, ESP32-S3, FreeRTOS, sensor integration, Bluetooth Low Energy (BLE), MQTT, MediaPipe
Standards Familiarity: ISO 14971:2019, ISO/TR 24971:2020, and IEC 62304
Language: Indonesian (Native), English (Upper-Intermediate)